Ewan John Dennis

AI/ML Engineer | Software Developer

+91 7994687722 | ewanjohndennis@gmail.com | linkedin.com/in/ewanjohndennis | github.com/ewanjohndennis | Kochi, Kerala

SUMMARY

Computer Science Engineering student with hands-on experience building agentic AI systems, ML pipelines, and data-intensive backend systems. Shipped a production multi-agent market intelligence platform during my Infosys Springboard internship, built a content decay scoring model at FlyRank AI against a 175K-page warehouse, and contributed to national-level hackathons including Kavach 2024. Work well independently, iterate quickly, and care about systems that hold up outside controlled environments.

EXPERIENCE

FlyRank AI — Jul 2026 – Aug 2026

Machine Learning Intern — Remote

- Built a content refresh prioritization model (Applied Search Intelligence, Lane 2) ranking decay risk across 175,205 pages from FlyRank's search analytics warehouse, scored end-to-end with a 3-tier actionable output queue.
- Trained LightGBM and Random Forest classifiers on a future-window label (prior-period signals predicting 30-day impression decline); LightGBM achieved ROC-AUC 0.973 and PR-AUC 0.9997 under strict client-grouped 5-fold cross-validation, beating the rule baseline's ROC-AUC of 0.903.
- Enforced `GroupKFold` splits by `client_hash_id` to prevent domain-authority memorization; naive random splits inflated ROC-AUC by 0.008, confirming the grouped approach as the honest zero-shot evaluation.
- Delivered a calibrated action playbook: IMMEDIATE_REFRESH (24%), SCHEDULED_UPDATE (21%), MONITOR (55%) – with reason codes and no-go automation rules for live publishing. Stack: Python, scikit-learn, LightGBM, DuckDB, pandas, Hugging Face datasets.
- **Paper:** https://ewanjohndennis-flyrankml.vercel.app/

Infosys Springboard (Virtual Internship 6.0) — Feb 2026 – Apr 2026

AI Engineer Intern — Remote

- Designed and deployed a multi-agent AI architecture with three parallel agents running concurrently via `ThreadPoolExecutor`, reducing data-collection latency by up to 3x compared to sequential execution.
- Built a synthesis pipeline integrating a FAISS vector database for RAG, a Groq LLM for report generation, and a PyTorch LSTM for time-series forecasting, running end-to-end with no manual steps.
- Iterated on prompt engineering pipelines across Azure OpenAI, HuggingFace, OpenRouter, and Groq — evaluating output quality and reliability under real-world rate limits and tracking provider performance across runs.
- Shipped a role-based Streamlit UI with MongoDB Atlas persistence, Gmail SMTP delivery, and PDF reporting via ReportLab. Generated 20+ intelligence reports in production.

PROJECTS

RTIIS | Real-Time Industry Insight & Strategic Intelligence System — 2026

- Flagship project from Infosys Springboard Virtual Internship 6.0. Multi-agent platform for automated market intelligence: three parallel agents collect news, competitor data, and financial signals concurrently via `ThreadPoolExecutor`, cutting data-collection latency by up to 3x.
- Synthesis agent uses FAISS for RAG, Groq for report generation, and a PyTorch LSTM for stock trend forecasting. Deployed on HuggingFace Spaces with MongoDB Atlas persistence, Gmail SMTP delivery, and PDF export via ReportLab. Supports up to 4 competitor companies per run.
- **GitHub:** github.com/Ewanjohndennis/RealTimeMarketIntelli

Kavach | AI Chargeback Decisioning System | Razorpay AI Buildathon — 2026

- Built an end-to-end chargeback decisioning system for Visa Reason Code 10.4 disputes as a solo entry. An XGBoost classifier estimates P(win) per dispute; a deterministic policy layer then routes to AUTO_CONTEST, MANUAL_REVIEW, or AUTO_ACCEPT based on expected financial value ($EV = P(win) \times amount - (1 - P(win)) \times$ contest fee).

- Used SHAP tree explainability to surface the strongest contributing features per prediction, kept strictly separate from representment evidence. Generated structured, evidence-grounded representment drafts for AUTO_CONTEST cases without LLM hallucination – all factual claims bound to supplied merchant telemetry.
- Exposed a FastAPI webhook API with SQLite persistence, a terminal operator interface for manual-review cases, and a held-out financial evaluation that reports net recovery and ROI against a contest-everything baseline. Stack: XGBoost, SHAP, FastAPI, SQLite, pandas.
- **GitHub:** github.com/Ewanjohndennis/Kavach

TrainCLI | End-to-End ML Training CLI 2025–2026

- Published to PyPI. Trains scikit-learn and XGBoost models directly from a CSV file with a single command, covering preprocessing, training, evaluation, and export. XGBoost is a lazy conditional dependency to keep the base install lightweight.
- Ships with an MCP server component, exposing training and inference as tools callable by any MCP-compatible agent.
- **GitHub:** github.com/Ewanjohndennis/traincli

Huddle | Real-Time Campus Collaboration Platform 2026

- Real-time platform for students to find and join active study sessions on campus. Validated real-time sync under concurrent sessions with sub-100ms Firestore latency; auto-cleans expired sessions. **Awarded 2nd Prize at Techsprint (GDGC MEC)**, from ∼40 teams, selected into a 10-team final.
- **GitHub:** github.com/Ewanjohndennis/huddle **Live:** huddlechats.vercel.app

SKILLS

AI / ML: Transformer Architecture, Agentic Workflows, Tool Calling, Embedding Models, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, LLM Integration, Context Engineering, FAISS Vector Database, Vector Embeddings, PyTorch, LSTM Forecasting, NLP, Fine-Tuning, Model Evaluation, MLOps, Data Preprocessing, scikit-learn, XGBoost, DuckDB
Providers & SDKs: Groq, HuggingFace Inference API, OpenRouter, Azure OpenAI, Anthropic SDK
Backend: Python, Node.js, Express.js, Spring Boot, REST API, PostgreSQL, MongoDB, Docker, FastAPI
Frontend: JavaScript (ES6+), TypeScript, React.js, Next.js, Tailwind CSS, Firebase
DevOps / Cloud: Git, CI/CD, Google Cloud Run, Vercel, Streamlit Cloud, HuggingFace Spaces
Methodologies: Agile (Scrum), Data Structures & Algorithms

EDUCATION

Model Engineering College 2024 – Present
B.Tech in Computer Science and Engineering — CGPA: 8.73 Kochi, Kerala

Vidyodaya School, Thevakkal 2022 – 2024
Senior Secondary (Class 12) — Percentage: 93% CBSE